

Progress:

I have three datasets all of them from AO3

Tags dataset:

This dataset is available on AO3 website that contains all the possible tags and their type, id and count which I will be referring to as fandom size.

This dataset had three missing tag names, so I dropped them. I also dropped merger_id column as it is mostly nan values, and I have no use for it.

I plotted the amount of works in media tags, highlighting that "Movie" has the most amount of works. Please note that in media I don't know whether the numbers given in millions or not as it wasn't stated in the page.

We also find that most works are rated "Teen and Up" with roughly 2.3 million works.

I also visualized top 10 fandoms based on the amount of work in each fandom.

Works Metadata with no text:

This dataset contains metadata about random sample of work from AO3.

This dataset is clean with no missing values as it was available on Kaggle.

I visualized the distribution of the numeric variables in the dataset while it is not normally distributed I plan to use ratios.

I done some feature engineering

- Kudos to hits ratio (what is the percentage of people that viewed the fan work that give it a kudos)
- Comments to hits ratio (what is the percentage of people that viewed the fan work that left a comment)
- Bookmarks to hits ratio (what is the percentage of people that viewed the fan work that bookmarked the fanwork)
- Fandom size: the amount of work in a fandom
- Hits to fandom size ratio to know how popular a fan work in its own fandom, I dropped fandoms with less than 100 to not bias our findings (a very small fandom <100 will have works that have views more than the actual size of the fandom cause the size here represent the amount of works not how many people are active in the fandom)
- I converted fan works word count to a fic_size categories: one chapters works are oneshot, 'short_multiple': (0 to 85000 words), 'medium': (85000 to 250000 words), 'long': (250000 to 2500000 words)

Text dataset:

The dataset I found that contain text and some of the metadata (missing kudos, comments, bookmarks and hits) is very large we are talking about millions of fan works I loaded only two files around 150,000 works.

Data cleaning

I dropped 356 rows that were nan in some columns, I checked the percentage of missing values in each column to see if I will drop the column, drop rows or find a solution to replace the nan values. I dropped columns with missing values above 80%, as for Relationship, Characters, and Additional Tags columns nan means the author choose not to include them so I didn't drop their missing values I changed it to N/A, I dropped missing values in Fandom, Archive Warning and Category as it is a small percentage.

I also converted Chapters and Words columns into int columns.

I choose to keep fan works that are text and in English.

The final dataset shape was (125271, 21)